

Higher Order ODEs

1 Linear ODEs

1.1 General Theory

Theorem 1.1 (Existence and Uniqueness Theorem).

Let $I \subset \mathbb{R}$ be an open interval and suppose functions $g, p_0, p_1, \dots, p_{n-1}$ are continuous on I . Then for $t_0 \in I$ and real numbers $y_0, y_1, \dots, y_{n-1}$, there exists exactly one solution to the IVP

$$\begin{cases} y^{(n)} + p_{n-1}(t)y^{(n-1)} + \dots + p_1(t)y' + p_0(t)y = g(t), \\ y(t_0) = y_0, \\ y'(t_0) = y_1, \\ \vdots \\ y^{(n-1)}(t_0) = y_{n-1}, \end{cases}$$

Theorem 1.2 (Principle of Superposition).

If $y_1, y_2, \dots, y_n$ are solutions to the homogeneous linear ODE

$$y^{(n)} + p_{n-1}(t)y^{(n-1)} + \dots + p_1(t)y' + p_0(t)y = 0,$$

then any linear combination

$$y = c_1y_1 + c_2y_2 + \dots + c_ny_n,$$

where $c_1, c_2, \dots, c_n$ are constants, is also a solution to the homogeneous equation.

Definition 1.1 (Wronskian).

The Wronskian of n functions $y_1, y_2, \dots, y_n$ is defined as the determinant

$$W(y_1, y_2, \dots, y_n)(t) = \begin{vmatrix} y_1(t) & y_2(t) & \dots & y_n(t) \\ y_1'(t) & y_2'(t) & \dots & y_n'(t) \\ \vdots & \vdots & \ddots & \vdots \\ y_1^{(n-1)}(t) & y_2^{(n-1)}(t) & \dots & y_n^{(n-1)}(t) \end{vmatrix}.$$

Definition 1.2 (Fundamental Set of Solutions).

The set of n solutions $y_1, y_2, \dots, y_n$ to an n -th order homogeneous linear ODE defined on an interval I is called a **fundamental set of solutions (FSS)** if every solution to the homogeneous equation can be expressed as a linear combination of $y_1, y_2, \dots, y_n$.

Theorem 1.3.

If $p_0, \dots, p_{n-1}$ are continuous on an interval I and $y_1, \dots, y_n$ are solutions to the homogeneous linear ODE

$$y^{(n)} + p_{n-1}(t)y^{(n-1)} + \dots + p_1(t)y' + p_0(t)y = 0,$$

then every solution to the ODE can be expressed as a linear combination of $y_1, \dots, y_n$ if and only if

$$\exists t \in I \text{ s.t. } W(y_1, \dots, y_n)(t) \neq 0$$

Theorem 1.4.

Let $y_1, \dots, y_n$ be the solutions to the homogeneous linear ODE

$$y^{(n)} + p_{n-1}(t)y^{(n-1)} + \dots + p_1(t)y' + p_0(t)y = 0, \quad \text{for } t \in I.$$

Then

$$y_1, \dots, y_n \text{ are linearly independent} \iff \exists t \in I \text{ s.t. } W(y_1, \dots, y_n)(t) \neq 0.$$

Theorem 1.5 (Abel's Theorem).

If $y_1, \dots, y_n$ are solutions to the homogeneous linear ODE

$$y^{(n)} + p_{n-1}(t)y^{(n-1)} + p_{n-2}(t)y^{(n-2)} + \dots + p_1(t)y' + p_0(t)y = 0, \quad \text{for } t \in I,$$

then the Wronskian is given by

$$W(y_1, \dots, y_n)(t) = c \cdot \exp\left(-\int p_{n-1}(t) dt\right) \quad \text{where } c \text{ is a constant depending on } y_1, \dots, y_n.$$

1.2 Homogeneous Equations with Constant Coefficients

For constants $a_n \neq 0, a_{n-1}, \dots, a_1, a_0 \in \mathbb{R}$,

$$a_n y^{(n)} + a_{n-1} y^{(n-1)} + \dots + a_1 y' + a_0 y = 0$$

Characteristic polynomial:

$$\begin{aligned} Z(r) &= a_n r^n + a_{n-1} r^{n-1} + \dots + a_1 r + a_0 \\ &= a_n (r - r_1)(r - r_2) \dots (r - r_n) \\ &= a_n (r - r_1)^{m_1} (r - r_2)^{m_2} \dots (r - r_k)^{m_k} \end{aligned}$$

where $r_1, r_2, \dots, r_n \in \mathbb{C}$ are the roots of $Z(r)$.

Complex-valued solutions

Complex roots must appear in conjugate pairs, i.e., if $r = \lambda + i\mu$ is a root, then its conjugate $\bar{r} = \lambda - i\mu$ is also a root. The complex-valued solution pair $y = e^{(\lambda+i\mu)t}$ and $y = e^{(\lambda-i\mu)t}$ can be re-expressed as the real-valued solution pair:

$$y = e^{\lambda t} \cos(\mu t) \quad \text{and} \quad y = e^{\lambda t} \sin(\mu t).$$

1.3 Non-Homogeneous Equations

1.4 Method of Undetermined Coefficients

1.5 Variation of Parameters